

Day 2: Environment Setup, Baseline Repositories & Data Models

Key Milestone: Development Environment & Core Database Schema

Sprint Focus: Execution of core deliverables for Day 2 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed: FastAPI Base Skeleton, React Boilerplate with Routing, Initial SQLAlchemy Migration Scripts.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership: Explainable AI, Intent & Phishing Detection, Social Engineering signals | **Key Modules:** NLP/AI Engine, Explainability, Content Risk scoring

Day 2 Execution: Implemented heuristic keyword matching engine and baseline Bayesian text classifier for urgency and psychological coercion indicators.
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership: Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis | **Key Modules:** Header Forensics, Mail Path, Infrastructure Risk

Day 2 Execution: Constructed raw MIME message parser in Python; created test suites with benign and forged RFC 822 email samples.
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership: URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection | **Key Modules:** IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 2 Execution: Built URL extraction regular expressions capable of decoding obfuscated, defanged, and redirected hyperlinked URIs.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership: Attack Relationship Graph, Similarity detection, Campaign grouping & correlation | **Key Modules:** Graph Engine, Campaign Detection, Similar Attack Detection

Day 2 Execution: Implemented Jaccard similarity and string Levenshtein distance modules to compare attacker-controlled subject lines and body templates.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership: Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions | **Key Modules:** Risk Scoring, Case Management, Response Recommendations

Day 2 Execution: Designed SQLAlchemy async models for Cases, Indicators, Artifacts, and Audit Logs; set up initial SQLite/Postgres migration scripts.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership: Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX | **Key Modules:** Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 2 Execution: Bootstrapped Vite + React frontend repository; configured Tailwind/Vanilla CSS design system with high-contrast cybersecurity theme.

System Integration & Quality Verification

Cross-Module Integration: Successfully connected frontend prototype to mock FastAPI endpoints; verified bidirectional JSON schema validation.

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 2 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 2 MILESTONE SIGNED OFF